

Den2Obj: A command-line tool for producing isosurfaces from electron density data files

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Summary

Electron density is a foundational quantity in electronic structure calculations. It describes the spatial distribution of electrons in molecules and materials, and is used to derive key properties such as total energy, electrostatic potential, and atomic forces. Because electron density is a three-dimensional scalar field, specialized visualization techniques, including contour plotting, isosurface generation, and volumetric rendering, are needed to interpret bonding, reactivity, and electron redistribution.

Den2Obj is a C++ command-line program for generating isosurfaces from electron density scalar fields. It supports input formats commonly used in electronic structure calculations, including CHGCAR and PARCHG files produced by VASP, as well as Gaussian Cube files. The resulting isosurfaces can be exported in widely supported geometry file formats, including STL (Stereolithography), PLY (Polygon File Format), and OBJ (Wavefront), enabling integration with visualization, rendering, and post-processing tools. In addition, Den2Obj can convert scalar fields to a documented native .d2o format, a lossless binary container that stores the grid and unit-cell metadata while selecting an efficient compression method automatically.

Statement of need

Isosurfaces play a central role in scientific research and engineering applications by providing an effective means of visualizing complex scalar fields and understanding intricate physical phenomena. An isosurface represents a set of points in a field where a scalar quantity, known as the *isovalue*, remains constant. Many software packages support isosurface generation, including Open Data Explorer (Thompson et al., 2004), MATLAB (The MathWorks Inc., 2022), ParaView (Ahrens et al., 2005), VESTA (Momma & Izumi, 2011), PyVista (Sullivan & Kaszynski, 2019), and chemistry-oriented tools built on such libraries, such as ChemVista (ChemParse, 2025). These packages and libraries are valuable for interactive inspection, scripting, and application-specific visualization workflows.

Den2Obj occupies a narrower niche: reproducible, non-interactive conversion of electronic-structure grid data into mesh and volume assets that can be used in automated analysis, documentation, and rendering pipelines. It directly reads formats commonly produced by VASP (Hafner, 2008), such as CHGCAR and PARCHG, as well as Gaussian (Frisch et al., 2016) Cube files. The resulting isosurfaces can be exported in widely used 3D geometry formats, including Stereolithography (.stl), Polygon File Format (.ply), and Wavefront (.obj).

Isosurface generation in Den2Obj is carried out using either the marching cubes algorithm (Lorensen & Cline, 1987) or the marching tetrahedra algorithm (Burke, 1994), both of which can be selected via command line arguments. These algorithms are implemented with OpenMP parallelization to leverage the performance of modern multi-core CPUs. When visualizing

41 wavefunctions rather than electron densities, it is often useful to generate separate isosurfaces
42 for positive and negative lobes. Den2Obj accommodates this by enabling dual isosurface
43 generation through a single command line argument.

44 An illustrative example is presented in Figure 1, which displays the canonical molecular orbitals
45 of the benzene molecule. These orbitals were computed using the PyQInt program (Filot, 2025).
46 Isosurfaces were generated via the marching cubes algorithm as implemented in Den2Obj,
47 producing .ply files that were subsequently imported into Blender (*Blender - a 3D Modelling
48 and Rendering Package*, 2018), along with the atomic coordinates of benzene, for rendering.

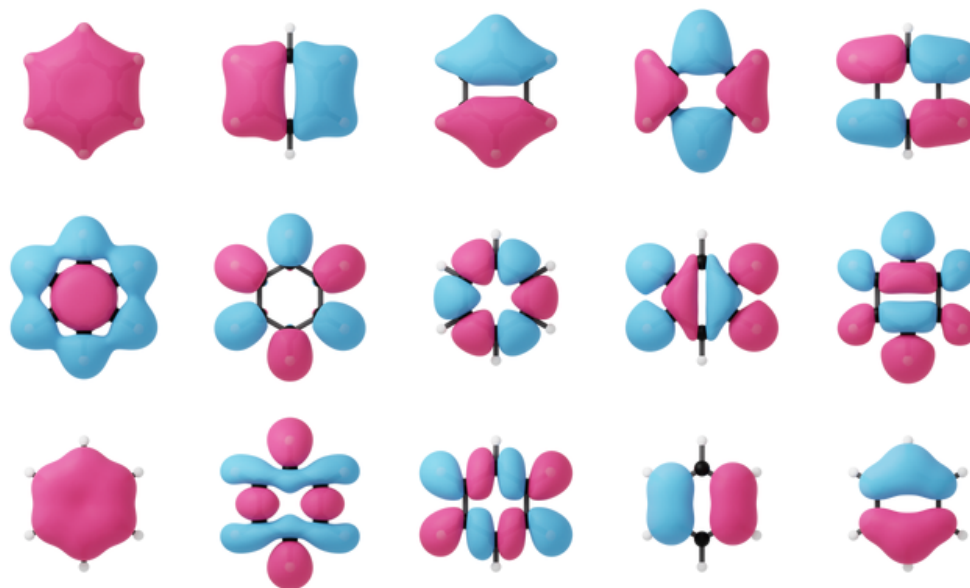


Figure 1: Isosurfaces of the first 15 canonical valence molecular orbitals of benzene.

49 For efficient research data management purposes, Den2Obj is also able to convert CHGCAR and
50 PARCHG files to its own custom .d2o format. This format is one of the distinctive components
51 of the software: it provides a documented, lossless binary representation of dense scalar
52 fields, including unit-cell and grid metadata, so that large electronic-structure outputs can be
53 archived and reloaded without reparsing text-based CHGCAR or PARCHG files. Upon conversion,
54 the program explores multiple compression algorithms, namely gzip, lzma, bzip2, Zstandard
55 (zstd) and Blosc, and uses the one that yields optimal results. These options cover Lempel-
56 Ziv and Huffman-style compression (Huffman, 1952; Ziv & Lempel, 1977), block-sorting
57 compression based on the Burrows-Wheeler transform (Fenwick, 1996), the Zstandard format
58 (Collet & Kucherawy, 2021), and cache-aware block compression for numerical data (Alted,
59 2010). Such size reductions are expected when converting human-readable grid data to
60 compressed binary storage; nevertheless, .d2o provides a convenient archival and intermediate
61 representation, with files that can be as small as about 10% of the original CHGCAR or PARCHG
62 file size.

63 Another distinguishing feature is direct conversion of electronic-structure scalar fields to
64 OpenVDB (Museth, 2013) files for volumetric rendering in programs such as Blender. In
65 contrast to the rendering of isosurfaces, the main advantage of volumetric rendering is that
66 internal details and density variations are more prominently shown, providing a comprehensive
67 and nuanced understanding of the scalar field. In a way, volumetric rendering lies in between
68 isosurfaces and contour plots in terms of visualizing a scalar field. An example for the molecular
69 orbitals of benzene is provided in Figure 2. For demonstration and testing purposes, also a
70 scalar field generator functionality is included that can create a number of relevant scalar fields

71 to test the algorithms on.

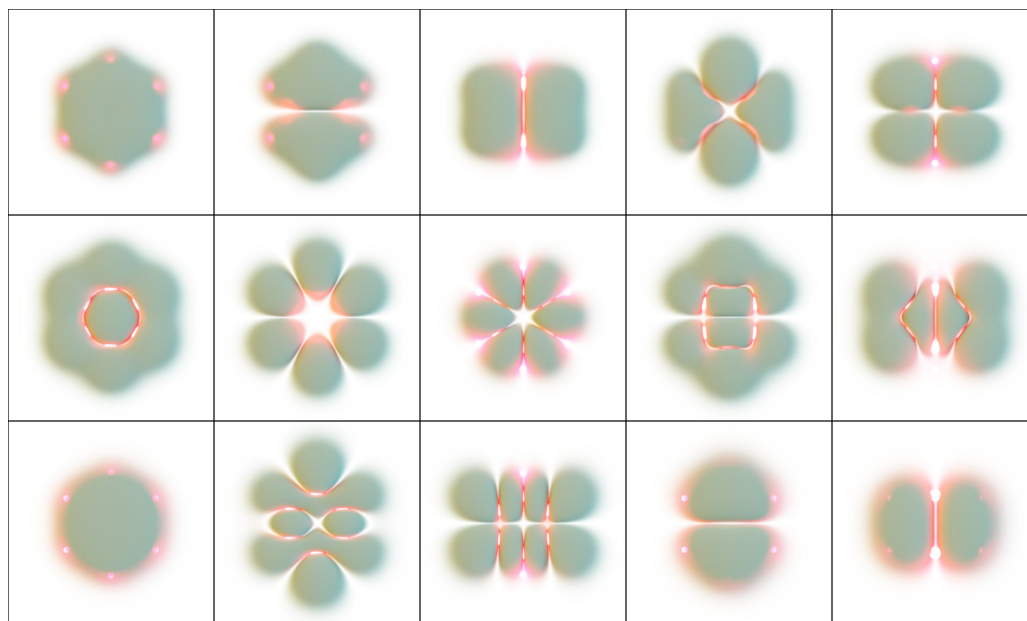


Figure 2: Volumetric rendering of the electron density associated with the first 15 canonical valence molecular orbitals of benzene using the OpenVDB format.

72 Den2Obj requires a relatively small set of dependencies: Eigen3 (Guennebaud et al., 2010),
73 Boost (Koranne, 2011), TCLAP (Smoot et al., 2009), and compression libraries for gzip, lzma,
74 bzip2, Zstandard and Blosc. Creation of VDB files requires the presence of the OpenVDB
75 library (Mueeth, 2013). The user can select during compilation whether they want to include
76 this functionality or not. Den2Obj is designed to be used by researchers and students working
77 in computational materials modelling and quantum chemistry. The present package builds on
78 an earlier command-line implementation of the same workflow. That earlier implementation
79 was used in the studies reported in (Filot et al., 2016; Su et al., 2016, 2018).

80 An extensive user guide including examples, compilation instructions, tutorials (including a
81 rendering tutorial in Blender) and documentation of the command-line arguments, is available
82 at <https://den2obj.imc-tue.nl/>.

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